
Topic 11

Activity Planning

Activity planning is a crucial aspect of project management, particularly in software development. It helps ensure that the project progresses smoothly and is completed on time, within scope, and on budget. Below are the detailed notes on **Work Breakdown Structure (WBS)** and **Scheduling Techniques** in software project management.

1. Work Breakdown Structure (WBS)

Definition:

A **Work Breakdown Structure (WBS)** is a hierarchical decomposition of a project into smaller, more manageable components or tasks. It is a project management tool used to break down the scope of work into smaller, detailed activities that can be assigned, monitored, and controlled.

Purpose of WBS:

- It helps in organizing the project scope.
- It clarifies project deliverables.
- It defines each task and its associated resources.
- It provides a framework for scheduling, cost estimation, and risk management.

WBS Levels:

- **Level 1:** Represents the overall project or objective.
- **Level 2:** Major deliverables or phases of the project.
- **Level 3:** Sub-deliverables or smaller tasks.
- **Level 4 and beyond:** Further breakdown of tasks, typically to a point where they can be easily assigned to a team member.

Example of WBS for a Software Project: Let's consider a software development project for a **Mobile Application**.

- **Level 1:** Mobile Application Development Project
 - **Level 2:** Planning Phase
 - **Level 3:** Requirements Gathering
 - **Level 3:** Feasibility Study
 - **Level 2:** Design Phase
 - **Level 3:** UI/UX Design
 - **Level 3:** System Architecture
 - **Level 2:** Development Phase
 - **Level 3:** Front-End Development
 - **Level 3:** Back-End Development

- **Level 2: Testing Phase**
 - **Level 3: Unit Testing**
 - **Level 3: Integration Testing**
 - **Level 2: Deployment Phase**
 - **Level 3: App Store Submission**
 - **Level 3: Production Deployment**
-

2. Scheduling Techniques in Software Project Management

Effective scheduling is essential to ensure that the project is completed within the specified timeframe. Below are common **scheduling techniques** used in software project management.

2.1. Gantt Chart

A **Gantt chart** is a visual representation of the project schedule. It shows the tasks to be completed on the vertical axis and time on the horizontal axis. Each task is represented by a horizontal bar, which shows its start date, duration, and end date.

Example: For the Mobile Application project, a Gantt chart might show the following:

- **Requirements Gathering** (January 1st to January 10th)
- **UI/UX Design** (January 11th to January 25th)
- **Front-End Development** (February 1st to March 10th)

2.2. Critical Path Method (CPM)

Critical Path Method (CPM) is a scheduling technique used to determine the longest sequence of tasks (the critical path) in the project. This path determines the minimum project duration. Tasks on the critical path cannot be delayed without affecting the project's completion date.

Example: In the Mobile Application project, the critical path could be:

- Requirements Gathering → UI/UX Design → Front-End Development → Back-End Development

2.3. Program Evaluation and Review Technique (PERT)

The **PERT** technique is a statistical tool used to analyze and represent the tasks involved in completing a project. PERT uses three estimates for each task:

- **Optimistic time (O):** The minimum possible time.
- **Pessimistic time (P):** The maximum possible time.
- **Most likely time (M):** The best guess for the time required.

The expected time for a task can be calculated using the formula:

$$E(T) = O + 4M + P / 6$$

Example: For the task of **UI/UX Design**, assume the following:

- Optimistic time: 5 days
- Most likely time: 10 days
- Pessimistic time: 15 days

Using the formula:

$$E(T) = 5 + 4(10) + 15 / 6 = 10 \text{ days}$$

2.4. Resource Levelling

Resource levelling is a technique used to resolve resource conflicts in the schedule. It involves adjusting the project schedule to ensure that resources are used efficiently without overloading them.

Example: If a developer is assigned to two tasks at the same time, resource levelling will adjust the schedule to ensure that the developer works on one task before moving to the next.

2.5. Time Boxing

Time boxing is a technique where a fixed amount of time is allocated to a specific activity. Once the time box is reached, the task is either completed or postponed, and the team moves on to the next activity.

Example: A team might allocate 2 weeks to complete the **UI/UX Design** phase. Even if the design is not fully completed, the team will move on to the development phase after 2 weeks.

Scenario-Based Questions with Solutions

1. **Scenario 1:** *Your project is behind schedule, and the critical path is delayed due to the Front-End Development phase. How can you address this using scheduling techniques?*

Solution: To address the delay, you could:

- Use **resource levelling** to ensure developers are focused on the most critical tasks without overloading them.
 - Consider using **time boxing** to ensure the Front-End Development phase stays within the set timeline, even if the task isn't fully completed.
 - If possible, split the task into smaller parallel tasks to **fast track** the development.
2. **Scenario 2:** *You are tasked with estimating the time required to complete the Back-End Development phase of a project. What scheduling technique would you use and why?*

Solution: You can use **PERT** to estimate the time. By gathering the optimistic, most likely, and pessimistic time estimates from the team members, you can calculate the

expected time to complete the Back-End Development phase. This helps in managing uncertainties and gives a realistic estimate.

3. **Scenario 3:** *You are managing a project with several interdependent tasks. The testing phase can only start after the development phase is completed. How can you ensure that there are no delays using CPM?*

Solution: By using **Critical Path Method (CPM)**, you can identify the critical path of the project and focus on the most critical tasks that affect the project's completion. By ensuring that the development phase stays on track, the testing phase will start as scheduled. Any delay in the development phase will directly impact the overall project completion.

وَقُلْ رَبِّ زِدْنِي عِلْمًا

PUACP